



RU

2021

1 (a)

- ▼ Define with example (i) Population and Sample, (ii) Parameter and Statistic **#revised-2**

(i) Population and Sample:

Population refers to the entire set of individuals, objects, or events that we are interested in studying and drawing conclusions about. It is often denoted by the letter 'N'.

Example: Let's say we want to study the average height of all students in a university. The population, in this case, would be all the students studying at that particular university.

Sample is a smaller portion of the population. It is often denoted by the letter 'n'.

Example: In the same scenario, instead of measuring the height of every student, we randomly select a group of, say, 100 students from the university and measure their heights. This group of 100 students would be considered the sample.

(ii) Parameter and Statistic:

A parameter is a numerical value that describes a specific characteristic of the entire population. Common examples are population mean (μ), population standard deviation (σ), etc.

Example: If we want to know the average age of all citizens in a country, the actual average age of the entire population would be the parameter.

Statistic is a numerical value that describes a specific characteristic of a sample. Common examples are sample mean ($\bar{x}$), sample standard deviation (s), etc.

Example: If we randomly select a group of 500 citizens from that country and calculate the average age of this sample, the result would be the statistic.

1 (b) 3

- ▼ Define Chi-square sampling distribution **#2016**. Mention some important properties **#2018 #2017** and uses of Chi-square distribution **#revised-2**

Chi-Square Sampling Distribution:

The chi-square (χ^2) sampling distribution is a probability distribution that arises from the chi-square statistic, which is commonly used in statistical hypothesis testing and confidence interval construction.

Properties of Chi-Square Distribution:

1. The shape of the chi-square distribution depends on the degrees of freedom (df).
2. All values of the chi-square random variable are non-negative.
3. It is not bounded
4. It takes values from 0 to positive infinity.

Uses of Chi-Square Distribution:

1. Compares observed data with expected predictions
2. Compares categorical data from different groups

3. Creates confidence intervals for variance
4. Checks if variances are equal between groups

1(c)

▼ Find MGF and CGF of Chi-square distribution. Hence find mean variance, β_1 and β_2 #2017 #revised-2

MGF of Chi-square distribution:

Moment Generating Function (MGF) of Chi-square (χ^2) Distribution :

We know,
P.d.f of χ^2 is, $f(\chi^2) = \frac{e^{-\frac{\chi^2}{2}} \cdot (\chi^2)^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})}$; $0 < \chi^2 < \infty$

Now, MGF of χ^2 distribution,

$$M_{\chi^2}(t) = E[e^{t\chi^2}] = \int_0^{\infty} e^{t\chi^2} f(\chi^2) d\chi^2$$

Expectation of

$$= \int_0^{\infty} e^{t\chi^2} \frac{e^{-\frac{\chi^2}{2}} \cdot (\chi^2)^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} d\chi^2$$

$$= \int_0^{\infty} \frac{e^{-\chi^2(\frac{1}{2}-t)}}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} d\chi^2$$

$$\begin{aligned}
 &= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^\infty e^{-x^2(\frac{1}{2}-t)} (x^2)^{\frac{n}{2}-1} dx^2 \\
 &= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^\infty e^{-ax} x^{\lambda-1} dx = \frac{\Gamma(\lambda)}{a^\lambda} \\
 &= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \frac{\Gamma(\frac{n}{2})}{(\frac{1}{2}-t)^{\frac{n}{2}}} \\
 &= \frac{1}{2^{\frac{n}{2}}} \times \frac{1}{(\frac{1-2t}{2})^{\frac{n}{2}}} = (1-2t)^{-n/2}
 \end{aligned}$$

CFG of Chi-square distribution, also finding mean variance, β_1 and β_2 :

Find the Cumulant Generating Function (CGF) of Chi-square distribution, Find β_1, β_2 , Mean and Variance.

The MGF of (χ^2) distⁿ with n df is given by,

$$M_{\chi^2}(t) = (1-2t)^{-\frac{n}{2}}$$

$$\therefore \text{CGF}, K_{\chi^2}(t) = \log\{M_{\chi^2}(t)\}$$

$$\Rightarrow K_{\chi^2}(t) = \log(1-2t)^{-\frac{n}{2}}$$

$$\Rightarrow K_{\chi^2}(t) = -\frac{n}{2} \log(1-2t) \quad [\log x^n = n \log x]$$

$$\Rightarrow K_{\chi^2}(t) = -\frac{n}{2} \left[-2t - \frac{(2t)^2}{2} - \frac{(2t)^3}{3} - \frac{(2t)^4}{4} - \frac{(2t)^5}{5} - \dots \right]$$

$$[\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots]$$

$$\Rightarrow K_{\chi^2}(t) = nt + \frac{n}{2} \cdot \frac{4t^2}{2} + \frac{n}{2} \cdot \frac{8t^3}{3} + \frac{n}{2} \cdot \frac{16t^4}{4} + \frac{n}{2} \cdot \frac{32t^5}{5} + \dots$$

$$\Rightarrow K_{\chi^2}(t) = nt + nt^2 + \frac{4}{3}nt^3 + 2nt^4 + \frac{32n}{10}t^5 + \dots$$

Equating on both sides we get, w.r.to t we get,

$$1 + k_1(t) + k_2 \frac{t^2}{2!} + k_3 \frac{t^3}{3!} + k_4 \frac{t^4}{4!} = n t + n t^2 + \frac{4}{3} n t^3 + 2 n t^4$$

Now,

$$k_1(t) = n t$$

$$\Rightarrow k_1 = n = \mu'_1 = \text{mean}$$

$$k_2 \frac{t^2}{2!} = (n t^2)$$

$$\Rightarrow \frac{k_2}{2!} = n$$

$$\Rightarrow k_2 = n \cdot 2! = 2n$$

$$\Rightarrow k_2 = 2n \quad [2! = 2 \times 1 = 2] = \mu'_2 = \text{variance}$$

$$k_3 \frac{t^3}{3!} = \frac{4}{3} n t^3$$

$$\Rightarrow \frac{k^3}{3!} = \frac{4}{3} n$$

$$\Rightarrow \frac{k^3}{3 \times 2 \times 1} = \frac{4}{3} n$$

$$\Rightarrow k_3 = \frac{6 \times 4}{3} n$$

$$\Rightarrow k_3 = 8n$$

$$k_4 \frac{t^4}{4!} = 2n t^4$$

$$\Rightarrow \frac{k_4}{4 \times 3 \times 2 \times 1} = 2n$$

$$\Rightarrow k_4 = 48n$$

$$\begin{aligned}
 \therefore \mu_4 &= 4\mu_1 + 3\mu_2^2 = (4) \cdot 2n + 3(2n)^2 \\
 &= 48n + 12n^2 \quad [\mu_2 = 2n] \\
 \therefore \beta_1 &= \frac{\mu_3}{\mu_2^3} = \frac{(8n)^2}{(2n)^3} = \frac{64n^2}{8n^3} = \frac{8}{n} \\
 \therefore \beta_2 &= \frac{\mu_4}{\mu_2^2} = \frac{48n + 12n^2}{(2n)^2} \\
 &= \frac{48n}{4n^2} + \frac{12n^2}{4n^2} \\
 &= \frac{12}{n} + 3
 \end{aligned}$$

2 (a)

▼ Define with examples (i) Unbiased estimate, (ii) Consistent estimate **#revised-2**

(i) Unbiased Estimate:

An estimate is considered unbiased if, on average, it equals the true value of the parameter being estimated. In other words, the expected value of the estimate is equal to the true value of the parameter.

Example

To estimate the average age of all students in a class, we randomly selected five students and found their ages to be 20, 22, 21, 20, and 23 years. The average of these ages (21.2 years) is an unbiased estimate of the average age of all students in the class if, on average, the sample mean is equal to the true average age.

(ii) Consistent Estimate:

A consistent estimate is one that becomes increasingly accurate as the sample size (or number of observations) grows larger. *In other words, as we gather more data and increase the sample size, the estimate approaches the true population parameter.*

Example

To estimate the probability of getting heads in a fair coin toss, starting by flipping the coin ten times and getting heads four times. The estimate is $4/10 = 0.4$ (40% chance of getting heads).

To get a more accurate estimate, we keep flipping the coin, for hundred times, and we get heads 50 times. The new estimate is $50/100 = 0.5$ (50% chance of getting heads).

So, as we keep increasing the number of coin flips, the estimate will get closer and closer to the true probability of 0.5

2 (b)

- ▼ What do you mean by sufficient estimator? What is Factorization theorem? **#revised-2**

An estimator of a parameter θ which gives as much information about θ as is possible from the sample at hand is called a sufficient estimator. Sufficient estimators exist when one can reduce the dimensionality of the observed data without loss of information.

Factorization Theorem (Neymann)

$T = t(x)$ is sufficient for θ if and only if the joint density function L of the sample values can be expressed as

$$L = g(T, \theta) h(x)$$

Where

$g(T, \theta)$ depends of θ and x only through value of $T = t(x)$

$h(x)$ is independent of θ

2 (c)

- ▼ Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ population. Find sufficient estimator for μ and σ^2 **#revised-2**

Example: Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ population.
Find sufficient estimators for μ and σ^2 .

Solution: Define the likelihood function

$$\begin{aligned} L &= \prod_{i=1}^n f(x_i, \theta) \\ &= \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\} \\ &= \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \left(\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2 \right) \right\} \end{aligned}$$

Which is of the form $g(T, \theta)h(x)$

Where $h(x) = 1$

Here, $T = (t_1(x), t_2(x))$
 $= (\sum x, \sum x^2)$

Thus, $\sum x$ is the sufficient estimator for μ .

And $\sum x^2$ is sufficient estimator for σ^2

3 (a)

▼ What is point estimation? What are the properties of a good estimator? #revised-2

Point estimation is a method in statistics used to estimate an unknown parameter of a population based on sample data.

Properties of a Good Estimator:

1. Has a small variance, providing precise and accurate estimates.
2. Contains all necessary information for unbiased inferences.
3. Becomes more accurate as sample size increases
4. Small **Mean Squared Error (MSE)**, combining low bias and low variance.
5. Gives estimates equal to the true parameter value on average.

3 (b) 2.75

▼ Mention some properties of maximum likelihood function #revised-2

1. becomes more accurate as the sample size increases.
2. depends only on the data through sufficient statistics.
3. normally distributed in large samples.
4. performs well in large samples, even with imperfect assumptions.
5. consistently estimates the true parameter value in large samples.
6. approaches the lowest possible variance for unbiased estimators.

3 (c)

▼ Suppose $x_1, x_2, \dots, x_n$ be a random sample of size n from poisson distribution with parameter θ . Obtain the MLE of θ and show that the estimator is unbiased #2015

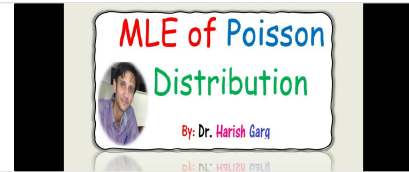
MLE of Poisson Distribution in 4 minutes

This video explains #MLE of #Poisson distribution.

Other videos @DrHarishGarg

Sampling Distribution: <https://youtu.be/Cd14ahGJG58>

▶ <https://www.youtube.com/watch?v=Ty94V1MqPMM>

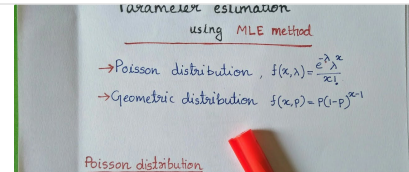


ML estimate of Poisson and Geometric distribution

#mleexample #poissondistribution #geometricdistribution

mle or maximum likelihood estimation is a method to estimate the parameter of a statistical model by maximizing the likelihood function. This tutorial describes the parameter estimation of poisson

▶ <https://www.youtube.com/watch?v=NAkNRTbVOWk>



Maximum Likelihood Estimation (MLE) Poisson Distribution in 2020 (8) | Statistics Tutor

In this video, we have discussed and solved most important problem of Maximum Likelihood Estimate for Poisson Distribution with parameter 'lambda' and also we have told about method of moments using conceptual method. So, must watch this video till end for more gain concepts.

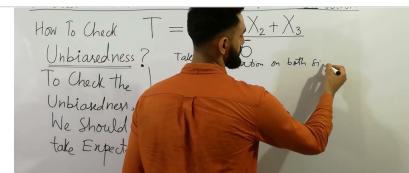
▶ <https://www.youtube.com/watch?v=BbJtyl9ubq0>



Unbiased Estimator Problem With Solution in 2022

In 2022, In this video, I have explained that how to check the unbiasedness and how to solve the problems of unbiased estimators with an easiest and shortcut method which nobody tells on YouTube and Google. This is first video about unbiased estimator problems that will clear your all concepts.

▶ <https://www.youtube.com/watch?v=CIPXkmXitnk>



4 (a)

- ▼ What do you mean by contingency table? #2018 #2016 #2015 What are the uses of such table? #revised-2

Contingency Table:

A contingency table, is a tabular representation used to display the relationship between two categorical variables.

Uses of Contingency Table:

1. Summarizing Categorical Data
2. Comparing Group Characteristics
3. Quality Control for Defect Analysis
4. Studying Social Phenomena and Behaviors
5. Hypothesis Testing for Independence

4 (b)

For a 2×2 contingency table prove that Chi-square test of independence gives #2018 #2016 #2015 #revised-2

$$X^2 = \frac{N(ad - bc)^2}{(a + c)(a + b)(c + d)(b + d)}, N = a + b + c + d$$

- ▼ Answer

Chi-Square Test for 2x2 Contingency Table

2x2 contingency Table $(a+b)(c+d) - (b+c+d+a)a =$

A \ B	B ₁	B ₂	Row Total (RT)
A ₁	a	b	a+b
A ₂	c	d	c+d
Column Total (CT)	a+c	b+d	N=a+b+c+d

$$\frac{ad-bc}{N}$$

We know that,

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = \sum_{i=1}^n \frac{(f_o - f_e)^2}{f_e}$$

O_i = observed frequency

$$\therefore E(a) = \frac{(a+b)(a+c)}{N}$$

$$E_i = \text{expected} = \frac{1}{(a+b)} + \frac{1}{(c+d)} + \dots + \frac{1}{(b+d)} \therefore E(b) = \frac{(a+b)(b+d)}{N}$$

O_i	E_i
a	$E(a) = \frac{(a+b)(a+c)}{N}$
b	$E(b) = \frac{(a+b)(b+d)}{N}$
c	$E(c) = \frac{(c+d)(a+c)}{N}$
d	$E(d) = \frac{(c+d)(b+d)}{N}$

$$\chi^2 = \frac{1}{(b+d)(c+d)} + \dots$$

Now, $\chi^2 = \frac{[a - E(a)]^2}{E(a)} + \frac{[b - E(b)]^2}{E(b)} + \frac{[c - E(c)]^2}{E(c)} + \frac{[d - E(d)]^2}{E(d)}$ — (1)

$a - E(a) = a - \frac{(a+b)(a+c)}{N}$

$= \frac{a(a+b+c+d) - (a+b)(a+c)}{N}$

$= \frac{\cancel{a^2} + \cancel{ab} + \cancel{ac} + ad - \cancel{a^2} - \cancel{ac} - \cancel{ba} - bc}{N}$

$= \frac{ad - bc}{N}$

$d + 0$	d	a
$b + 0$	b	c
$b + d + n = N$	$b + d$	$a + c$

Similarly we will get,

$$b - E(b) = \frac{ad-bc}{N} \quad c - E(c) = \frac{ad-bc}{N} \quad d - E(d) = \frac{ad-bc}{N}$$

Substituting these values in (1),

$$\chi^2 = \frac{(ad-bc)^2}{N^2} \left[\frac{1}{E(a)} + \frac{1}{E(b)} + \frac{1}{E(c)} + \frac{1}{E(d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad-bc)^2}{N^2} \left[\frac{N}{E(a)} + \frac{N}{E(b)} + \frac{N}{E(c)} + \frac{N}{E(d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad-bc)^2}{N^2} \left[\frac{1}{(a+b)(a+c)} + \frac{1}{(a+b)(b+d)} + \frac{1}{(a+c)(c+d)} + \frac{1}{(b+d)(c+d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad-bc)^2}{N} \left[\frac{a+b+c+d}{(a+b)(a+c)(b+d)} + \frac{a+b+c+d}{(a+c)(c+d)(b+d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad-bc)^2}{N} \left[\frac{1}{(a+b)(a+c)(b+d)} + \frac{1}{(a+c)(c+d)(b+d)} \right]$$

$$\Rightarrow \chi^2 = (ad-bc)^2 \frac{a+b+c+d}{(a+b)(a+c)(c+d)(b+d)}$$

$$\Rightarrow \chi^2 = \frac{N(ad-bc)^2}{(a+b)(a+c)(c+d)(b+d)}$$

4 (c)

In an experiment on immunization of cattle from tuberculosis, the following results are obtained

	Affected	Not affected
Inoculated	10	35
Not inoculated	13	6

Examine whether vaccination control diseases using the critical value 3.29 at $\alpha = 0.05$

▼ Answer

5 (a)

▼ Explain Type-I and Type-II errors #2015 #2014 #revised-1

Type I Error (False Positive):

A Type I error occurs when we reject a true null hypothesis. Type I error is denoted by the symbol α (alpha) and is referred to as the significance level of the test. Lowering the significance level reduces the probability of Type I error but increases the probability of Type II error.

Type II Error (False Negative):

A Type II error occurs when we fail to reject a false null hypothesis. Type II error is denoted by the symbol β (beta) and represents the probability of correctly rejecting a false null hypothesis.

5 (b)

▼ What is power of a test? #2018 #2017 Explain the different steps to construct statistical test of hypothesis #2015 #revised-1

The power of a statistical test, denoted as $1 - \beta$ (beta), is the probability of correctly rejecting a false null hypothesis.

Steps to Construct a Statistical Test of Hypothesis:

1. **State the Null and Alternative Hypotheses:** Clearly state the null hypothesis (H_0) and the alternative hypothesis (H_a). The null hypothesis represents no effect, while the alternative hypothesis represents the effect we want to test.
2. **Choose the Test Statistic:** Select an appropriate test statistic that measures the difference between the sample data and the null hypothesis under the assumptions of the statistical test.
3. **Set the Significance Level (α):** Determine the significance level (α), which is the probability of making a Type I error. Commonly used significance levels are 0.05 and 0.01.
4. **Calculate the Critical Value or P-value:** Based on the chosen significance level and the sampling distribution of the test statistic, calculate the p-value associated with the test statistic.
5. **Establish the Rejection Region:** If the test statistic falls in the rejection region (determined by the critical value) or if the p-value is less than the chosen significance level (α), reject the null hypothesis.
6. **Compute the Power of the Test ($1 - \beta$):** If the alternative hypothesis is true, calculate the power of the test.
7. **Interpretation and Conclusion:** Based on the test results, interpret the findings and draw conclusions regarding the hypothesis test.

5 (c)

▼ Jeffrey, as an eight-year-old, established an average time of 16.43 seconds for swimming the 25-yard freestyle, with a standard deviation of 0.8 seconds. His dad, Frank, thought that Jeffrey could swim the 25-yard freestyle faster by using goggles. Frank bought Jeffrey a new pair of expensive goggles and timed Jeffrey for 15 25-yard freestyle swims. For the

15 swims, Jeffrey's average time was 16 seconds. Frank thought that the swim goggles helped Jeffrey to swim faster than the 16.43 seconds. Conduct a hypothesis test using a preset $\alpha = 0.05$. Assume that the swim times for the 25-yard freestyle are normal

Answer

Set up the Hypothesis Test:

Since the problem is about a mean, this is a **test of a single population mean**.

$$H_0 : \mu = 16.43, H_a : \mu < 16.43$$

For Jeffrey to swim faster, his time will be less than 16.43 seconds. The "<" tells you this is left-tailed.

Determine the distribution needed:

Random variable: $\bar{X}$ = the mean time to swim the 25-yard freestyle.

Distribution for the test: $\bar{X}$ is normal (population standard deviation is known: $\sigma = 0.8$)

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) \text{ Therefore, } \bar{X} \sim N\left(16.43, \frac{0.8}{\sqrt{15}}\right)$$

$\mu = 16.43$ comes from H_0 and not the data. $\sigma = 0.8$, and $n = 15$.

Calculate the p - value using the normal distribution for a mean:

$p\text{-value} = P(\bar{x} < 16) = 0.0187$ where the sample mean in the problem is given as 16.

$p\text{-value} = 0.0187$ (This is called the **actual level of significance**.) The p - value is the area to the left of the sample mean is given as 16.

Graph:

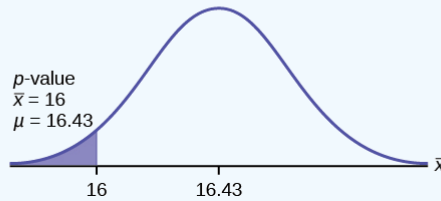


Figure 8.2.7

$\mu = 16.43$ comes from H_0 . Our assumption is $\mu = 16.43$.

Interpretation of the p - value: If H_0 is true, there is a 0.0187 probability (1.87%) that Jeffrey's mean time to swim the 25-yard freestyle is 16 seconds or less. Because a 1.87% chance is small, the mean time of 16 seconds or less is unlikely to have happened randomly. It is a rare event.

Compare α and the p - value:

$$\alpha = 0.05, p\text{-value} = 0.0187, \alpha > p\text{-value}$$

Make a decision: Since $\alpha > p\text{-value}$, reject H_0 .

This means that you reject $\mu = 16.43$. In other words, you do not think Jeffrey swims the 25-yard freestyle in 16.43 seconds but faster with the new goggles.

Conclusion: At the 5% significance level, we conclude that Jeffrey swims faster using the new goggles. The sample data show there is sufficient evidence that Jeffrey's mean time to swim the 25-yard freestyle is less than 16.43 seconds.

6 (a)

- ▼ What is Non-parametric test of hypothesis? #2016 How does it differ from parametric test #2017 #revised-1

Non-parametric tests of hypothesis is a statistical test that do not make any assumptions about the underlying distribution of the population.

Criteria	Non-parametric Test	Parametric Test
Assumptions	No assumptions about the data	Assumptions about data distribution
Efficiency	Less powerful	More powerful
Sample Size Requirement	Larger sample sizes needed	Smaller sample sizes can be used
Test Statistics	Rank-based or frequency-based	Based on means, variances, etc.

6 (b)

- ▼ Explain the advantages and disadvantages of Non-parametric test over parametric test **#revised-1**

Advantages of Non-parametric Tests over Parametric Tests:

1. do not rely on assumptions about the underlying data distribution
2. It is effective when dealing with skewed data
3. It can handle ordinal data
4. less affected by extreme outliers
5. It perform well with small sample sizes

Disadvantages of Non-parametric Tests compared to Parametric Tests:

1. less powerful than their parametric counterparts, especially when the data does conform to parametric assumptions.
2. By converting data to categories, non-parametric tests may lose some information present in the original data
3. have fewer variations available compared to parametric tests

6 (c)

- ▼ Discuss about median test stating usual assumption

The median test is a non-parametric statistical test used to determine whether there is a significant difference between the medians of two or more groups.

Assumptions of the Median Test:

1. The observations in each group are independent of each other.
2. The samples collected from each group are obtained randomly from their respective populations
3. Each observation can belong to only one category or group.
4. The test does not require data to be measured on a continuous scale.

Steps in Performing the Median Test:

1. Set up the null and alternative hypotheses. The null hypothesis (H_0) assumes that the medians of all groups are equal, while the alternative hypothesis (H_a) suggests that at least one median is different.
2. Combine the data from all groups and rank them in ascending order.
3. The test statistic for the median test follows a chi-square distribution. Formula: $\chi^2 = \sum [(O - E)^2 / E]$
4. The degrees of freedom (df) for the chi-square distribution are calculated as $(r - 1)$
5. Using the chi-square distribution table determine the critical value associated with the test statistic.
6. If the p-value is less than the chosen significance level (α), reject the null hypothesis and If the p-value is greater than α , fail to reject the null hypothesis

2020

1 (a) 2.75

- ▼ What do you mean by simple random sampling? Illustrate with an example

Simple random sampling is a method of selecting a sample from a larger population in such a way that every individual or item in the population has an equal chance of being included in the sample.

Illustration with an Example:

Suppose we want to conduct a survey to estimate the average age of students in a university. The university has a total of 1,000 students. To perform a simple random sample, we would follow these steps:

1. Assign a unique identifier (such as a number or a code) to each student in the university
2. Using a random selection process to choose a specific number of students for the sample
3. Select the students corresponding to those numbers from the list

1 (b)

▼ Suppose we select a random sample of size n from a population and have independent random variables $X_1, X_2, \dots, X_n$. Do those random variables have the same distribution? Explain

Yes, the random variables $X_1, X_2, \dots, X_n$ selected from a random sample of size n from a population have the same distribution. This is because they are independently and randomly sampled from the same population, which means that each random variable is subject to the same underlying distribution of the population.

1 (c)

▼ Suppose that of six students, the first student has \$1, the second student has \$2, the third student has \$3, the fourth student has \$4, the fifth student has \$5, and the sixth student has \$6. Consider the 1,2,3,4,5,6 dollar as the population. A sample of two students is selected. What is the probability that the sample mean is equal to \$3.5? write down every step of your calculation

Step 1: $nCr = n! / (r! * (n-r)!)$

Here, $n = 6$ and $r = 2$

$6C2 = 15$

15 possible samples = {1, 2}, {1, 3}, {1, 4}, {1, 5}, {1, 6}, {2, 3}, {2, 4}, {2, 5}, {2, 6}, {3, 4}, {3, 5}, {3, 6}, {4, 5}, {4, 6}, {5, 6}

Step 2: Sample Mean = (Sum of values in the sample) / (Number of items in the sample)

Sample Means: {1.5, 2, 2.5, 3, 3.5, 2.5, 3, 3.5, 4, 3.5, 4, 4.5, 4.5, 5, 5.5}

Step 3: There are two samples with a mean of 3.5: {2, 5} and {3, 4}

Step 4: Probability (P) = (Number of favorable outcomes) / (Total number of possible outcomes).

$P(\text{sample mean} = 3.5) = 2/15 \approx 0.1333$ (or approximately 13.33%)

2 (a)

▼ Write some names of statistics that can be used as estimator of the population

Mean, Median, Mode, Variance, Standard Deviation, Sample Proportion, Sample Correlation Coefficient, Sample Regression Coefficients, Percentiles.

2 (b)

▼ What are the criteria for good estimator? #2017 #2015

Unbiasedness, Efficiency, Consistency, Sufficiency, Robustness.

2 (c)

▼ Is 'sample mean' a good estimator in terms of unbiasedness? Explain

We can prove it mathematically,

Sample Mean is Unbiased Estimator of Population Mean

Population mean μ is an estimator of

We suppose that sample mean $\bar{x}$ is an estimator of population mean μ

We know that,
$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Taking expectation,

$$E(\bar{x}) = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right]$$

$$= \frac{1}{n} E\left[\sum_{i=1}^n x_i\right]$$

$$= \frac{1}{n} E[x_1 + x_2 + \dots + x_n]$$

$$\left[\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n \right]$$

$$= \frac{1}{n} [E(x_1) + E(x_2) + \dots + E(x_n)]$$

$$= \frac{1}{n} [\mu + \mu + \dots + \mu]$$

$$= \frac{1}{n} [\mu(1 + 1 + \dots + 1)]$$

$$= \frac{1}{n} [\mu(n)]$$

$$E(\bar{x}) = \mu$$

This proves that the expected value of the sample mean ($\bar{x}$) is equal to the population mean (μ), which means the sample mean is a good estimator in terms of unbiasedness.

2 (d)

▼ Is 'median' a consistent estimator? Explain with example

Yes, the median is a consistent estimator.

Consistent estimator is the one that becomes increasingly accurate as the sample size grows larger.

For example, true population median of the following data: {1, 3, 5, 7, 9} is the middle value 5.

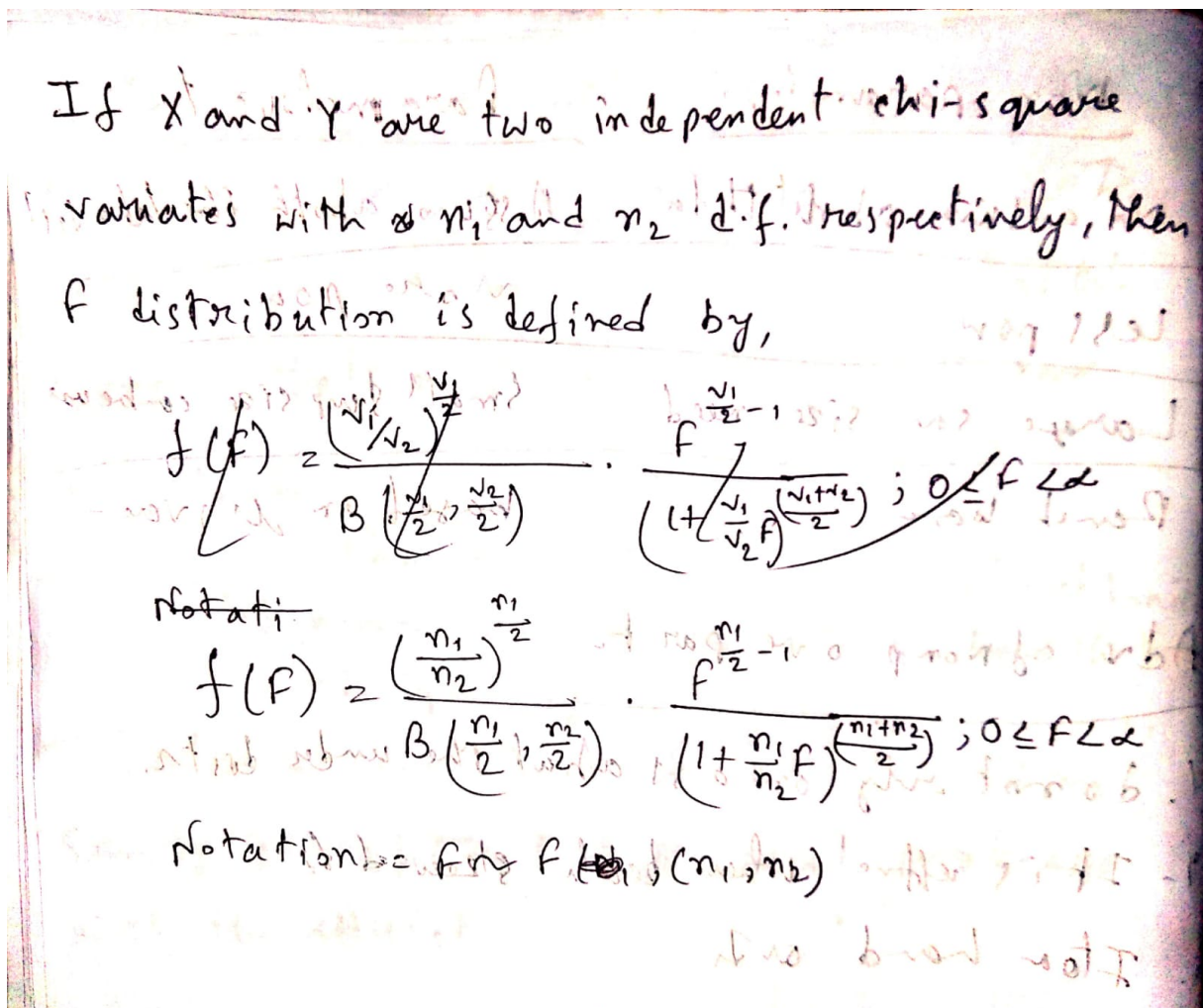
Now,

1. Sample size ($n = 1$):
Sample: {1}
Sample median = 1
2. Sample size ($n = 2$):
Sample: {1, 3}
Sample median = $(1 + 3) / 2 = 2$
3. Sample size ($n = 3$):
Sample: {1, 3, 5}
Sample median = 3
4. Sample size ($n = 4$):
Sample: {1, 3, 5, 7}
Sample median = $(3 + 5) / 2 = 4$
5. Sample size ($n = 5$):
Sample: {1, 3, 5, 7, 9}
Sample median = 5

As we increase the sample size, we get closer to the true population median of 5. This demonstrates that the median is a consistent estimator

3 (a)

▼ Define F-distribution with degrees of freedom n_1 and n_2 and point out its uses.



Uses of F-Distribution:

1. Tests equality of means in three or more groups.
2. Checks equal variances in groups.
3. Compares variances in different batches for consistency.
4. Tests overall significance of regression model.

3 (b)

▼ Discuss the relationship among F, t, and χ^2 distribution

Distribution	Definition	Parameter	Relationship
F-Distribution	Probability distribution of the F-statistic	Degrees of freedom: (df ₁ , df ₂)	- Derived from the ratio of two independent chi-square distributions. - Used in ANOVA and regression analysis for comparing variances and testing model significance.
t-Distribution	Probability distribution of the t-statistic	Degrees of freedom: df	- Derived from the standard normal distribution - Used in t-tests and confidence intervals for small sample sizes when population standard deviation is unknown.

χ^2 -Distribution	Probability distribution of the chi-square	Degrees of freedom: df	- Square of a standard normal distribution. - Used in chi-square tests for testing independence and goodness-of-fit.
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4 (a)

▼ What do you mean by tests of goodness of fit?

Tests of Goodness of Fit are statistical tests used to assess how well an observed frequency distribution fits an expected theoretical distribution. These tests determine whether the observed data follows a specific theoretical distribution or if there are significant differences between the observed and expected frequencies. They are commonly used in various fields, such as biology, economics, and social sciences

4 (b)

Suppose a coin is tested 60 times with the results given in the following table

Event	n_I
Heads	32
Tails	28

Is this a fair coin? The null hypothesis is

$$H_1 : n_I = np_I$$

where n_I is the number of observed frequencies and np_I is the expected frequencies in which p_I are the theoretical probabilities. Given that, $\alpha = 5\%$

▼ Answer

$$H_0 : n_I = np_I$$

$$H_1 : n_I \neq np_I$$

$$\alpha: 5\% (0.05)$$

$$np_I = n \times p_I = 60 \times 0.5 = 30$$

$$\chi^2 = \sum ((O_i - E_i)^2 / E_i)$$

$$\chi^2 = \sum \frac{(n_I - np_I)^2}{np_I} = \frac{(32 - 30)^2}{30} + \frac{(28 - 30)^2}{30} = \frac{4}{15}$$

$$df = 2 - 1 = 1$$

Calculate p value: p value not given

- If the test statistic $\chi^2 >$ critical value, reject the null hypothesis.
- If the P-value $< \alpha$, reject the null hypothesis.

4 (c)

▼ Which statistical test is suitable to test equality of variances? Explain

The statistical test suitable to test the equality of variances is the **F-test for equality of variances**

1. The F-test compares variances of different groups in ANOVA to check if they are equal or different.
2. It compares the ratio of larger variance to smaller variance, using the F-statistic.
3. Decision about variance equality is based on comparing the F-statistic to critical values from the F-distribution.

5 (a)

- ▼ Define (i) null hypothesis #2018 (ii) critical region #2018 #2014

(i) Null Hypothesis:

The null hypothesis, denoted as H_0 , is a statement of no effect between variables in a statistical hypothesis test. It represents the default assumption that there is no significant effect. The objective is to either reject or fail to reject the null hypothesis based on the evidence provided by the data.

(ii) Critical Region:

The critical region, is a range of values of the test statistic that leads to the rejection of the null hypothesis in a statistical hypothesis test. It is defined based on the chosen significance level (α), which is the probability of making a Type I error

5 (b)

- ▼ Describe how you would test the hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$

we will use the **F-test for equality of variances to test the hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$, where σ_1^2 and σ_2^2 are the variances of two populations**

1. Set Null and Alternative Hypothesis,

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

2. Set Significance Level for example, $\alpha = 0.05$

3. Calculate Test Statistic, $F = \text{Variance of Sample 1} / \text{Variance of Sample 2}$

4. Determine degrees of freedom for both df_1 and df_2

5. calculate the P-value associated with the F-statistic

6. if the P-value is less than the significance level (α), reject the null hypothesis or else fail to reject the null hypothesis.

5 (c)

- ▼ **The estimates of variances from two random samples of sizes 7 and 16 are 5.6 and 7.0 respectively. Test whether the samples could have arisen from populations with the same sample variance**

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_1 : \sigma_1^2 \neq \sigma_2^2$$

$$\alpha = ?$$

$$F = \text{Larger Sample Variance} / \text{Smaller Sample Variance}$$

$$F = 7.0 / 5.6 = 1.25$$

$$df_1 = 16 - 1 = 15 \text{ and } df_2 = 7 - 1 = 6$$

$$p\text{-value: ?}$$

P-value is less than the significance level (α), reject the null hypothesis or else fail to reject the null hypothesis.

6 (a)

- ▼ In what circumstances we need to perform non-parametric test?

1. When data is not normally distributed.
2. With small sample sizes.

3. For ordinal or categorical data.
4. Dealing with outliers or skewed data.
5. When variances between groups are heterogeneous.
6. Handling non-numeric data.
7. In the presence of assumption violations.

6 (b)

▼ **A particular shoe store believes that the median foot size of teenage boys is 10.25 inches. To test this hypothesis, the foot size of each of a random sample of 50 boys was determined. Suppose that 36 boys had sized in excess of 10.25 inches. Does this disprove the hypothesis that the median size is 10.25? Use sign test to solve this problem**

6 (c)

▼ Discuss the purposes of run test and rank-sum test

Run Test:

The run test is used to examine whether a sequence of data points follows a random pattern. The main purpose of the run test is to test the hypothesis of randomness in the data.

Rank-Sum Test (Mann-Whitney U Test):

The rank-sum test is a non-parametric test used to compare the distribution of two independent samples. Its main purpose is to determine whether the two samples come from the same population or if there are differences between them.

2019

1 (a)

What do you mean by sampling distribution? Write down the probability density function of Chi-square distribution of n degree of freedom (d.f)

1 (b)

If X has a Chi-square distribution with n d.f, find the MGF of Chi-square distribution

1 (c)

Write down important uses of Chi-square distribution

2 (a)

What is likelihood function?

2 (b) 3

What are the principles of maximum likelihood estimation method?

2 (c)

Find maximum likelihood estimator of parameters when n samples are drawn from Normal distribution

3 (a)

Define statistic, estimator and estimate with example

3 (b)

Describe three properties of a good estimator, in short

3 (c)

Show that sample average $\bar{x}$ is an unbiased estimator of population mean, but sample variance $S^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ is a biased estimator of population variance

4 (a)

Distinguish between:

1. Null hypothesis and alternative hypothesis
2. Type I error and Type II error

4 (b)

How does a t-test differ from a normal test?

4 (c)

A sample of 900 items taken from population with standard deviation 2.61 cms. The mean of the sample is 3.4 cms. Test whether the sample come from the population with mean 3.25 cms. To be noted $\Pr\{Z < -1.73\} = \Pr\{Z > 1.73\} = 0.089$

5 (a)

What is critical region? How the alternative hypothesis affects in computing critical region?

5 (b)

Describe different steps for testing hypothesis of equality of two proportions

5 (c)

Random samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. Among the respondents, 200 men and 325 women were in favor of the proposal. Test the hypothesis that proportions of men and women in favor of the proposal are same. To be noted $\Pr\{Z < -1.269\} = \Pr\{Z > 1.269\} = 0.10$

6 (a)

What is a contingency table? How does it differ from a correction table?

6 (b)

What is the null hypothesis in a test of independence? Define the expression for the value of Chi-square in a rxc contingency table

6 (c)

Two sample polls of votes for two candidates A and B for a public office are taken, one from among the residents of rural areas. The results are given in the adjoining table. Examine whether the nature of the area is related to voting preference in the election

	Votes for		
Area	A	B	Total
Rural	620	380	1000
Urban	550	450	1000
Total	1170	830	2000

Area	Votes for		Total
	A	B	
Rural	620	380	1000
Urban	550	450	1000
Total	1170	830	2000

[Value of the cumulative distribution function of Chi-square at 10.09 with 1 df is 0.0015]

2018

1 (a)

- ▼ Define chi-square (χ^2) variate and write down its pdf

1 (b)

- ▼ Show that χ^2 **distribution tends** to normal distribution for large degree of freedom

1 (c)

- ▼ Write down some important properties of χ^2 distribution #2021

2 (a)

- ▼ Define F-statistic. Write down the p.d.f of F-statistic

2 (b)

- ▼ If F follows F-statistic on (n_1, n_2) degrees of freedom and show that the statistics $(1 + \frac{n_1}{n_2} F)^{-1}$ has beta distribution

2 (c)

- ▼ State 2 (two) main applications of F-statistic

3 (a)

- ▼ What are the methods of point estimation?

3 (b)

▼ Define estimate and estimator with examples

3 (c)

Let $x_1, x_2, \dots, x_n$ be a random sample from a poisson distribution with pdf given by $f(x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}; x = 0, 1, 2, \dots$

Find the maximum likelihood estimator of λ . **Show** the estimator is unbiased.

▼ Answer

4 (a)

▼ What do you mean by statistical test of hypothesis? Define with examples (i) null hypothesis #2020 (ii) critical region #2020 and (iii) level of significance

4 (b)

▼ What is BCR? When does a test become MP test?

4 (c)

▼ A bulb manufacturing company claims that the average longevity of their bulb is 3.65 years with a standard deviation of 0.16 years. A random sample of 36 bulbs gave a mean longevity of 4.45 years. Does the sample mean justify the claim of the manufacturer? [Use 5% level of significance]

5 (a)

▼ What do you mean by the power of a test? #2021

5 (b)

▼ Describe how you will test the following hypothesis $H_0 : \rho_1 = \rho_2 = \dots = \rho_k (k > 2)$

5 (c)

A set of 8 correlation coefficients and the corresponding sample sizes are given below. Test the homogeneity of these coefficients

v	0.231	0.464	0.539	0.357	0.628	0.136	0.204
n	10	12	8	15	18	11	10

▼ Answer

6 (a)

What is contingency table? #2021 For a 2×2 contingency table #2021

a	b
c	d

show that,

$$\chi^2 = \frac{N(ad - bc)^2}{(a + c)(a + b)(c + d)(b + d)}, N = a + b + c + d$$

▼ Answer

2017

1 (a)

▼ Define chi-square (χ^2) variate and its p.d.f

1 (b)

▼ By using moment generating function (MGF) find β_1 and β_2 #2021

1 (c)

▼ Find the mode of χ^2 -distribution. Mention some important properties #2021 and application of χ^2 -distribution

2 (a)

▼ Define Student's t statistics. Formulate its sampling distribution

2 (b)

▼ Show that Student's t distribution reduces to the standard normal distribution for large degrees of freedom

3 (a)

▼ Define point estimator. What are the criteria of good estimator? #2020

3 (b)

▼ What do you mean by sufficient statistic? Let $x_1, x_2, \dots, x_n$ be a random sample from a poisson variate with parameter ' μ ' then show that $\bar{x}$ is a sufficient static for ' μ '

3 (c)

Let $x_1, x_2, \dots, x_n$ be a random sample of size ' n ' from normal distribution with p.d.f is

$$f(x, \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} - \alpha < x < \alpha, -\alpha < \mu < \alpha, \sigma^2 > 0$$

Find the MLE of ' μ ' and σ^2

▼ Answer

4 (a)

▼ Distinguish between Type 1 and Type 2 errors. Define: (i) Power of a test #2021, (ii) Level of significance and (iii) Degree of freedom. Describe the procedure for Testing of Hypothesis

4 (b)

▼ The coefficient of correlation obtained from a random sample of 20 pairs is 0.50. Test the population correlation coefficient ($\rho = 0$) at 5% level of significance. [$t_{0.05,18} = 2.10$]

5 (a)

When do you use independent samples t-test?

Researchers are interested in the mean level of some enzyme in each and compute a sample mean 22. It is known that the variable of interest is approximately normally distributed with a variance of 45. Can you conclude that the mean enzyme level in this population is different from 25 at the 5% level of significance? [$Z_{0.05} = 1.96$]

▼ Answer

5 (b)

▼ For a simple random sample of adults, IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. A simple random sample of 13 statistics professors yields a standard deviation of $s = 7.2$. Assume that IQ scores of statistics professors are normally distributed and use a 0.05 significance level to test the claim that $\sigma = 15$. [The tabulated value χ^2 with d.f. 12 at 5% level of significance are 4.404 and 23.337]

6 (a)

▼ Distinguish between parametric and non-parametric statistical tests #2021. Discuss the advantages and disadvantages of non-parametric test

6 (b)

▼ Derive sign test, stating clearly the assumptions made for small sample case

6 (c)

Use the sign test to see whether there is a difference between the number of days required to collect an account receivable before and after a new collection policy. Use the 0.05 significance level.

Before: 33 36 41 32 39 47 34 29 32 40 42

After: 35 29 38 34 37 47 36 32 30 34 41 38

▼ Answer

2016

1 (a)

▼ Define Chi-square (χ^2) distribution #2021. Show that χ^2 -distribution tends to normal distribution for large degrees of freedom

1 (b)

▼ State and prove additive property of χ^2 -distribution. If x has density function $f(x) = e^{-x}, x > 0$. Then show that $2x$ follows χ^2 -distribution with 2-degrees of freedom

2 (a)

▼ Define F variate. Find the mode of F distribution. If X has F distribution with n and m degrees of freedom. Mention some important properties of F distribution

3 (a)

▼ Define point estimation with example #2021. What are the methods of point estimation?

3 (b)

What is MLE? State and prove the invariance property of MLE. Let $x_1, x_2, \dots, x_n$ be a random sample from

$$f(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, 2, \dots, n.$$

Find the MLE of p.

▼ Answer

4 (a)

▼ What do you mean by statistical hypothesis? Distinguish between simple and composite hypotheses. Let a random sample of size n is drawn from a normal population with mean μ and known variance σ^2 . How would you test the hypothesis that mean is equal to μ_0 ?

4 (b)

▼ The average IQ of university female students in Bangladesh is suspected to be more than the average 110 for all students. A random sample of 64 female students yielded a sample average IQ of 115.5 and standard deviation of 20. Can you conclude that the average score of the female students is really more than 110? [$Z_{0.05} = 1.64$]

5 (a)

Define $r \times c$ contingency table. #2021 Show that in case of 2×2 contingency table, the test statistics becomes #2021

$$\chi^2 = \frac{N(ad - bc)^2}{(a + c)(a + b)(c + d)(b + d)},$$

Also mention Yate's correction for continuity

▼ Answer

5 (b)

▼ In a psychological test, 70 out of 100 boys came out successful while 60 out of 100 girls of the same age group as the boys passed the test. Do the data provide any evidence of difference in respect of abilities between the genders?

6 (a)

▼ What do you mean by non-parametric test? #2021 Discuss its importance. Describe the testing procedure of the run test

6 (b)

The following sequence is purported to be a set of random integers from 0 to 99. Use the run's test to test the hypothesis of the randomness at $\alpha = 0.05$ significance level. The sequence is
28,4,23,98,44,10,6,25,54,81,12,6,4,33,67,55,71,66,22,18,49,85

▼ Answer

2015

1 (a)

▼ Define Sampling Distribution with examples. Mention the names of sampling distributions which are frequently used

1 (b)

▼ Define χ^2 -distribution. Find moment generating function of χ^2 -distribution. Also find mean and variance of χ^2 -distribution

2 (a)

▼ State and prove t-distribution. Show that the odd ordered moments of t-distribution is zero. Write down the properties of t-distribution

2 (b)

▼ Show that t-distribution tends to normal distribution if the degrees of freedom tend to infinity

3 (a)

▼ What are the desirable criteria of a good estimator? #2020

3 (b)

Let $(x_1, x_2, \dots, x_n)$ be a random sample from a Poisson distribution with p.d.f given by

$$f(x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}; x = 0, 1, 2, \dots$$

Find the maximum likelihood estimator of λ . Show that the estimator is unbiased #2021

▼ Answer

4 (a)

▼ What do you mean by a statistical hypothesis? Describe different steps for testing statistical hypothesis. #2021 Write down the procedure to test the significance of regression coefficient

4 (b)

▼ A random sample of 10 persons is selected as follows: 5,2,0,4,16,14,10,11,6,8. Do you think that the average schooling year of the persons in population is 5? (Tabulated value at 5% with 9 d.f. is 2.26)

5 (a)

- ▼ Define Type I error #2021, Type II error #2021, Level of significance and Most powerful test.

5 (b)

- ▼ Suppose k random samples are drawn from normal population with mean $\mu_1, \mu_2, \dots, \mu_k$ and common variance σ^2 . Describe the procedure to test that the means are equal

6 (a)

- ▼ What is contingency table? #2021 What is form of χ^2 test statistic in case of a 2×2 contingency table? #2021

6 (b)

- ▼ For given information in the following table, test at level of significance 0.05 that whether level of education affects the job performance. [$\chi^2_{0.05,4} = 13.3$]

..... level of education affects the job performance. $\alpha = 0.05, 4 = 13.3$

Job Performance	Level of Education			
	Below primary	College	University	
Excellent	10	40	10	60
Good	30	30	20	80
Fair	10	30	20	60
	50	100	50	200

2014

1 (a)

- ▼ Define Chi-square statistic. Derive its sampling distribution

1 (b)

- ▼ State some uses of t-distribution. Describe how would you test the equality of variances of two normal populations

2 (a)

- ▼ Distinguish between MVB estimator and MV estimator. Show that MVB unbiased estimator is also sufficient

2 (b)

- ▼ Find the MVB unbiased estimator of variance of a normal population with mean $\mu = 0$. Also show that this estimator is also sufficient

3 (a)

- ▼ Define likelihood function. Describe the method of maximum likelihood

3 (b)

Let $(x_1, x_2, \dots, x_n)$ be a random sample from a normal distribution with p.d.f given by

$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < \infty, -\infty < \mu < \infty, \sigma^2 \geq 0$$

Find the maximum likelihood estimator of μ and σ^2

▼ Answer

4 (a)

▼ Define sample hypothesis and critical region #2020. Let $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_m$ be two random samples drawn from two normal populations $N(\mu_i, \sigma^2), i = 1, 2$ respectively. If σ^2 is unknown, how would you test null hypothesis that $H_0 : \mu_1 = \mu_2$

4 (b)

▼ Sample mean weight of 20 CSE students is 50kg and 10 ICE students is 45kg. If the sample variances of weights are 25 and 16, test whether the mean weights of CSE students is greater than the mean weights of ICE students. (Use $t_{0.05, 28} = 1.64$)

5 (a)

▼ Define type I error and type II error #2021. Describe the procedure of testing the hypothesis that population proportion is equal to a specified value p_0

5 (b)

▼ A nutritionist claims that 80 percent of the pre-school children in a certain country have protein-deficient diet. A sample survey reveals that it is true for 244 children out of 300. Is the nutritionist justified in his claim? Use a significant level of 0.01 [$Z_{0.01} = 2.33$]

6 (a)

▼ Describe the procedure of Fisher's exact test for testing the independence of two binary variables

6 (b)

For given information in the following table, test at level of significance 0.05 that whether class attendance affects the examination score

Class attendance	Examination score	
	A+	Less than A+
Less than 80%	3	6
80% and above	7	4

▼ Answer